Equilibrium 2034-Player Manual

0. Game Overview

A Rule of the Post-Collapse World: No Power Lasts Forever

In the year 2034, after the collapse of global civilization, essential resources have been exhausted.

The game follows a repeating loop:

Gamble → Win/Loss → Panic → Survive → Gamble

1. Nations and Infrastructure

Each nation depends on water to sustain a critical infrastructure.

Greenvale (Agriculture)

Represents China

-Key Infrastructure: Agriculture

Tidefall (Data Systems)

Represents the United States

-Key Infrastructure: Data Centers

Ironcrest (Nuclear Power)

Represents Russia

-Key Infrastructure: Nuclear Energy

2. Player Role

-Each player acts as a young envoy sent by their nation.

-In the post-collapse world, envoys hold greater power than old-era diplomats—but also bear greater responsibility.

-Your goal is simple:

·You must keep your nation alive for three rounds by maintaining a sustainable level of resources.

· After Round 3, the nation with the highest total resources is crowned

· The New Era Leader, guiding humanity out of the wasteland.

3. Outsider NPCs

Kael – Arms Dealer

・Controls the black market. Loyal only to resources.

・Pay resources to gain military power—or be ignored.

Lyra – Peace Mediator

・A former humanitarian worker.

・Attempts to preserve diplomacy and prevent total collapse.

Rorke – Intelligence Operative

・A war veteran haunted by loss.

・Provides espionage, sabotage, and hidden advantages.

4. Learning Goal

As a young member of a nation, players learn to recognize patterns of resource scarcity and exercise agency in complex geopolitical interactions, constantly choosing betw een high-risk self-interest and strategic cooperation to secure long-term survival and leadership in a post-collapse world.

5. Components

- Nation Cards (3)

Greenvale, Tidefall, Ironcrest

- Outsider NPC Cards (3)

Kael, Lyra, Rorke

- Event Cards

A mix of positive and negative events.

- Game Board

Displays infrastructures and resource pools.

Used for Gamble bets.

- Tokens

Resource Tokens

Used for betting, survival costs, diplomacy, and NPC actions.

- Military Power Tokens

Used to resolve conflicts through Kael.

- New Era Leader Trophy

Symbol of final victory.

- Powerful State Marker

Marks the nation with the highest resources in a round.

- Dice

・Gamble Die (D6)

Used for turn order, Gamble outcomes.

・Negotiation Die (D6)

Used to trigger forced negotiations.

6. Initial Setup

Number of Players: 3

Each player controls one nation.

Starting Resources:

Each nation starts each round with 1 resource.

Initial Resource Pools:

Roll the Gamble die three times:

・Roll 1–2: Add 1 resource to Pool A

・Roll 3–4: Add 1 resource to Pool B

・Roll 5–6: Add 1 resource to Pool C

This creates three shared resource pools.

7. Game Loop

Each round proceeds in the following order:

1.Gamble

2.Win/Loss

3.Panic

4.Survive

5.Next Round

Key Emotion: Desperation refers to a state where survival feels urgent—when resources are needed immediately or collapse is inevitable.

8. Gamble Action

Turn Order

All players roll the Gamble die.

Players act from lowest roll to highest roll.

Player Action

On your turn:

1.Choose a resource pool (A, B, or C).

2.Place 1 of your resources as a bet.

3.Roll the die.

Outcome

If the roll matches the number range of your chosen pool, you gain all resources in that pool, including your bet.

If the roll does not match, you gain resources from the pool corresponding to the number rolled.

9. Win and Loss Resolution

After all Gamble actions are completed, each nation counts its total resources.

Win

The nation with the highest resources becomes the Powerful State.

Loss

All other nations become Weaker States.

Win Effects

Greenvale (Agriculture)

May redraw one Event Card this round.

Ironcrest (Nuclear)

Draw two Event Cards, choose one to resolve.

Tidefall (Data)

After drawing a negative Event Card, may swap it with another nation.

Winning nations still resolve Event Cards. Victory does not remove risk.

Loss Effects

Greenvale – Food Gap

Trigger: Insufficient water for agriculture.

Effect: Draw two Event Cards.

Ironcrest – Reactor Overheating

Trigger: Insufficient cooling water.

Effect: Draw two Event Cards.

Tidefall – Data Shutdown

Trigger: Insufficient cooling water.

Effect: Draw two Event Cards.

Losing nations cannot independently obtain protection and must negotiate with winning nations.

10. Event Cards

Event Cards activate only during the Panic phase and last for one round.

Players draw Event Cards during the Win/Loss phase and may keep them secret.

Cards are single-use unless stated otherwise.

Positive Event Cards

Intelligence Privilege

Allows one free Rorke action (Eavesdrop, Sabotage, or Enhance).

Ceasefire Badge

Immediately ends a conflict without resolution.

Black Market Voucher

Next Kael trade converts 1 resource into 2 military power.

Prestige Badge

Forces another nation to ally with you for the round.

International Aid

Gain 1 immediate resource.

Negative Event Cards

Panic Surge

Lose 1 resource immediately.

Intelligence Jam

No Rorke actions this round.

Black Market Scam

First Kael trade this round loses 1 power.

Alliance Breakdown

All current alliances end immediately.

Global Reputation Loss

Cannot call allies this round.

Diplomacy Lockdown

No negotiations allowed; must escalate conflict.

11. Panic Phase

Mobilization (Weaker State)

If two weaker nations attempt mobilization, both roll dice.

Higher roll acts first.

To attack a Powerful State, the weaker nation pays 1 resource to initiate a clash.

Deterrence (Powerful State Response)

Concede

Pay 1 resource to Lyra to negotiate peace.

Escalate

Both sides purchase military power from Kael.

Each resource converts to equal power.

Combat Resolution

If the weaker state wins, it takes half of the powerful state's resources.

If the powerful state wins, it takes half of the weaker state's resources next round.

Call Ally

A weaker nation may request aid from another weaker nation.

Support

Resources are shared to purchase power.

Reject

The requesting nation loses 1 resource.

Rorke Actions

After power purchases, any nation may pay 1 resource to:

・Eavesdrop: View opponent's power investment.

・Sabotage: Secretly reduce opponent power by 1.

・Enhance: Secretly increase own power by 1.

12. Survive Phase

Each nation must pay its infrastructure survival cost:

Greenvale – Agriculture: 1 water

Tidefall – Data Centers: 2 water

Ironcrest – Nuclear Power: 3 water

If a nation cannot safely pay, it may request survival support.

Survival Negotiation

A nation in crisis proposes a deal to others, offering future advantages or concessions.

Negotiation Roll

All other nations roll a die.

The lowest roll must enter negotiation.

A binding deal must be formed, and agreed resources are paid immediately.

After survival resolution, the next round begins.

After Round 3, the nation with the highest resources is crowned

The New Era Leader.

Equilibrium 2034

Procedural Rhetoric, Systems Thinking, and Learning Through Constraint

1. Introduction

1.1 Problem: The Literacy – Reality Gap in Contemporary Education

Contemporary scholarship on literacy further complicates this gap by questioning the traditional alignment between learning and language-based instruction. Jewitt (2008) argues that contemporary classrooms operate within an increasingly multimodal communicational landscape, in which meaning is produced not solely through written or spoken language, but through configurations of image, action, sound, spatial arrangement, and interaction. From this perspective, how knowledge is represented is inseparable from how it is learned: the choice of mode shapes not only what counts as knowledge, but also the kinds of cognitive and social engagement that learning affords.

Despite these shifts, formal education continues to privilege print-based and linguistic forms of representation, often marginalizing experiential, procedural, and embodied modes of learning. This mismatch is particularly consequential for complex social systems—such as resource scarcity, infrastructural dependence, and geopolitical interdependence—which cannot be fully grasped through propositional explanation alone. As Jewitt (2008) notes, limiting learning to language-based literacy risks obscuring the systemic and relational dimensions of knowledge that emerge more clearly through multimodal engagement. This theoretical tension underscores the need for learning environments that allow students to interact with systems through action and constraint, rather than language alone.

1.2 Need: Why a Game-Based Intervention?

Traditional instructional approaches—such as lectures, readings, or case analyses—can raise awareness of global issues, but they often struggle to convey the experiential dimensions of uncertainty and scarcity that shape real-world decision-making. Serious games offer a distinctive pedagogical affordance by embedding constraints and consequences directly into interactive systems. Rather than being told how systems work, learners must act within them.

Empirical research on experiential and game-based learning suggests that simulations can support deeper understanding of complex systems by allowing learners to test strategies, encounter failure, and observe causal relationships over time. Multi-round gameplay enables iterative learning, where players gradually refine their understanding through trial, error, and reflection. Importantly, games can simulate

high-risk decision-making environments without real-world consequences, making them particularly suitable for exploring issues such as scarcity, cooperation, and power asymmetry.

Within this project, *Equilibrium 2034* adopts what Bogost (2008) terms **procedural rhetoric**—the idea that games persuade not through explicit statements, but through the logics embedded in their rules. Instead of instructing players what to think about resource inequality or geopolitical cooperation, the game is designed to let players *feel* these dynamics as they struggle to survive within a constrained system.

1.3 Learning Goal and Learning Group

Based on research in experiential and systems-oriented learning, *Equilibrium 2034* establishes the following learning goal:

> As a young member of a nation, the learner will be able to recognize patterns of resource scarcity and exercise strategic agency within complex geopolitical interactions—continuously choosing between high-risk self-interest and strategic cooperation—in order to ensure the survival and long-term sustainability of their nation in a post-collapse world.

This learning goal is operationalized through the game's turn-based survival system. Across multiple rounds, learners are expected to unpack several forms of knowledge: recognizing patterns of resource scarcity, understanding trade-offs between population and resources, making strategic judgments between cooperation and self-interest, and reflecting on how systems reward or punish particular behaviors.

The intended learning group for *Equilibrium 2034* is high school students enrolled in global studies or social studies courses. The game is designed as a synchronous classroom activity that supports embodied and contextualized learning. Rather than emphasizing rote memorization, the game encourages students to recall concepts, explain outcomes, and *perform* understanding through situated decision-making. In doing so, it aims to cultivate interdisciplinary literacy that connects social, political, and ethical reasoning.

2. Theoretical Framework and Game Design Mapping

2.1 Theory 1: Procedural Rhetoric and the *Survive* Action

Bogost's (2008) concept of procedural rhetoric provides the primary theoretical foundation for *Equilibrium 2034*. In this framework, game rules are not neutral constraints but expressive systems that articulate arguments through process. Rather than communicating meaning through text or images, procedural rhetoric persuades by requiring players to act within rule-based systems and experience their

consequences. In this sense, rules themselves function as arguments, and play becomes a form of interpretation.

The *Survive* action in *Equilibrium 2034* exemplifies this logic. Each nation must pay recurring infrastructure survival costs tied to its dominant system—agriculture, data centers, or nuclear energy. These systems are assigned different levels of water dependency: agricultural systems require the least water, data centers demand higher cooling resources, and nuclear systems impose the most severe requirements, with catastrophic consequences if unmet. These distinctions are not arbitrary balancing decisions. They encode a procedural claim about modern development: technological sophistication and national power often increase vulnerability under conditions of scarcity.

Crucially, players are not given the option to abandon infrastructure. Every nation must continuously pay survival costs, or else trigger famine, system overheating, or nuclear crisis. When a nation cannot meet these requirements, the game does not immediately end. Instead, it activates a *Survival Negotiation* mechanism, forcing the distressed nation to seek assistance from others. Aid is conditional and often requires political concessions, such as future turn disadvantages or the transfer of special privileges. Through this process, sovereignty and development capacity become negotiable assets.

The game further introduces forced negotiation rules that prevent complete neutrality during crises. Certain survival failures compel other nations to participate in negotiations, reflecting the spillover effects of infrastructural collapse in globally interconnected systems. Cooperation thus emerges not as a moral ideal, but as a structurally compelled response to systemic risk. Through repeated exposure to survival costs and negotiation pressure, players gradually recognize how development pathways, resource dependence, and structural inequality interact—precisely the form of experiential critique emphasized by procedural rhetoric.

2.2 Theory 2: Pattern Recognition and Iterative Learning

Koster's (2014) theory of fun frames learning as the process of pattern recognition. According to this view, games are engaging because they allow players to identify, test, and master patterns within systems. Learning occurs not through explanation, but through repeated interaction with feedback-rich environments.

Equilibrium 2034 operationalizes this theory through its cyclical gameplay structure: *Gamble → Win/Loss → Panic → Survive*. Across multiple rounds, players encounter recurring patterns related to resource volatility, NPC behavior, and infrastructure costs. Success depends less on memorizing rules than on recognizing how scarcity, risk, and cooperation interact over time.

Playtesting revealed that early strategies often prioritized short-term self-interest, leading to rapid collapse. Iterative adjustments to resource volatility and panic thresholds increased the salience of long-term consequences, prompting players to revise their approaches. Over time, players demonstrated improved anticipation of downstream effects and a growing capacity to balance immediate gains against systemic stability. Learning thus emerged through failure, adjustment, and reflection—consistent with Koster's account of pattern-based learning.

2.3 Theory 3: Gholdy Muhammad's Framework of Multidimensional Literacy

Gholdy Muhammad's (2020) framework expands the concept of literacy beyond technical skills to include **Identity, Skill, Intellect, Criticality, and Joy (Awe)**. This model emphasizes that meaningful learning must support sociopolitical understanding, identity formation, and emotional engagement alongside cognitive development.

In *Equilibrium 2034*, these dimensions are embedded within the game's resource systems and social interactions. Players assume the role of young envoys from one of three nations—Greenvale, Tidefall, or Ironcrest—each modeled with distinct infrastructure dependencies. These identities shape how players experience scarcity and risk, aligning with Muhammad's emphasis on identity as situated within broader sociopolitical systems.

The game cultivates *Skill* and *Intellect* through strategic decision-making across repeated cycles, requiring players to interpret resource distributions, negotiate alliances, and anticipate consequences. *Criticality* emerges as players encounter structural inequities embedded in the rules, such as asymmetrical panic effects and cascading disadvantages for weaker nations. These mechanics mirror real-world power imbalances, fostering awareness of how systemic conditions reproduce vulnerability and privilege independent of individual effort.

Finally, moments of *Joy* or *Awe* arise through shared agency—successful cooperation, narrowly avoided collapse, or collective survival under pressure. These emotional experiences support sustained engagement with complex ideas, reinforcing Muhammad's argument that affective investment is central to deep learning.

3. Playtesting Evidence and Iterative Refinement

The design of *Equilibrium 2034* evolved through multiple iterations. In early discussions with the instructor, the game's intended emotional core—"urgency"—was identified as insufficiently specific to drive player behavior. This feedback led to a reframing of the core affect as **desperation**, which more directly shaped decision-making under scarcity.

Initial classroom playtests revealed a mismatch between the stated learning goal of resource scarcity and overly generous resource conditions. In response, the system

was simplified to foreground scarcity as the central mechanic. Midway through development, technical constraints prompted a shift from a planned VR prototype to Twine. While this transition reduced sensory immersion, it clarified which mechanics were essential to the learning goal.

Ultimately, the team returned to a tabletop format, recognizing that transparent resource systems, face-to-face negotiation, and controllable rule structures best supported the intended learning outcomes. This iterative process reinforced a key design insight: educational effectiveness depends less on technological novelty than on alignment between mechanics, theory, and learning goals.

4. Conclusion and Implications

Equilibrium 2034 demonstrates how procedural rhetoric and multi-round strategic gameplay can support systemic understanding of resource scarcity, structural inequality, and geopolitical interdependence. By embedding arguments within rules rather than explanations, the game enables learners to experience complexity through action.

The project suggests that educational games can bridge the gap between classroom knowledge and real-world systems by transforming rules into curriculum. Learning emerges not through content transmission, but through structured experience. Future iterations may incorporate more dynamic events or culturally specific narratives to further support identity and critical awareness. Ultimately, *Equilibrium 2034* positions play as a mode of critical inquiry—one that allows learners to live within systems and learn from their consequences.

References

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